

Vishwavrinda Ayurveda Backend API And System Design

Generated for the Fastify + TypeScript WhatsApp AI backend.

Project Structure

text

src/

constants/ Shared API paths, pagination limits, collection names, indexes, and messages

lib/ Shared helpers for pagination, ObjectId validation, regex escaping, and request

body checks

parsers/ Meta WhatsApp webhook payload parsing

routes/ Fastify route registration and endpoint handlers

services/ Business logic, MongoDB access, OpenAI, WhatsApp, and knowledge file operations

types/ Shared TypeScript data models

System Design

text

Incoming WhatsApp Message

|

v

POST /webhook

|

v

src/parsers/whatsapp.ts

|

v

src/services/conversation.ts

|

+--> upsertUser() -----> MongoDB users

|

+--> createMessage(role=user) -----> MongoDB messages

|

+--> getKnowledge() -----> knowledge.md

|

+--> generateReply() -----> OpenAI Responses API

|

+--> AIResult type=message

| |

| v

| sendAndStoreMessage()

| |

| +--> sendMessages() -----> WhatsApp Cloud API

| |

| +--> createMessage(role=assistant) after successful send

|

+--> AIResult type=appointment

|

v

createAppointment() -----> MongoDB appointments

|

v

sendAndStoreMessage(server confirmation)

Data Storage

MongoDB collections:

- users
- messages
- appointments

File storage:

- knowledge.md remains file-based and is not stored in MongoDB.

Indexes created during database initialization:

- users: phoneNumber unique, lastActiveAt, firstSeenAt
- messages: phoneNumber + createdAt
- appointments: createdAt, isCompleted + createdAt, phoneNumber

Shared Rules

Pagination defaults:

- page = 1
- limit = 20
- maximum limit = 100

Paginated response format:

```
json
{
  "data": [],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 0,
    "totalPages": 0
  }
}
```

API Documentation

GET /

File:

- src/index.ts

Purpose:

- Health check endpoint.

Parameters:

- None.

Returns:

json

```
{  
  "status": "ok",  
  "service": "WhatsApp AI Bot"  
}
```

GET /webhook

Files:

- src/routes/webhook.ts
- src/config.ts

Purpose:

- Meta WhatsApp webhook verification.

Query parameters:

- hub.mode: expected subscribe
- hub.verify_token: must match VERIFY_TOKEN
- hub.challenge: returned to Meta when verification succeeds

Returns:

- 200 with the challenge string when valid.
- 403 with Verification failed when invalid.

POST /webhook

Files:

- src/routes/webhook.ts
- src/parsers/whatsapp.ts
- src/services/conversation.ts

Purpose:

- Receives incoming WhatsApp webhook events, parses the incoming message, and starts the AI conversation flow.

Body:

- Meta WhatsApp webhook payload.

Returns:

```
json  
{  
  "success": true  
}
```

Side effects:

- Creates or updates a user in users.
- Stores incoming user message in messages.
- Calls OpenAI.

- Sends WhatsApp reply.
- Stores assistant reply only after WhatsApp send succeeds.
- Creates appointment when AIResult returns a complete appointment.

GET /test-whatsapp

File:

- src/routes/webhook.ts

Purpose:

- Sends a fixed WhatsApp test message using the configured WhatsApp sender.

Parameters:

- None.

Returns:

```
json
{
  "success": true
}
```

GET /test-ai

Files:

- src/routes/webhook.ts
- src/services/knowledge.ts
- src/services/openai.ts

Purpose:

- Loads knowledge.md and asks the AI a fixed test question.

Parameters:

- None.

Returns:

```
json
{
  "success": true,
  "reply": {
    "type": "message",
    "reply": "string",
    "appointment": null
  }
}
```

GET /api/dashboard/stats

Files:

- src/routes/dashboard.ts
- src/services/dashboard.ts

Purpose:

- Returns dashboard counters calculated from existing MongoDB collections.

Parameters:

- None.

Returns:

```
json
{
  "totalUsers": 0,
  "newUsersThisWeek": 0,
  "newUsersThisMonth": 0,
  "totalMessages": 0,
  "totalAppointments": 0,
  "pendingAppointments": 0,
  "completedAppointments": 0,
  "appointmentsThisWeek": 0,
  "appointmentsThisMonth": 0
}
```

GET /api/appointments

Files:

- src/routes/appointments.ts
- src/services/appointments.ts

Purpose:

- Returns paginated appointment requests for dashboard management.

Query parameters:

- page: optional positive integer, default 1
- limit: optional positive integer, default 20, max 100
- status: optional pending, done, or all, default all
- search: optional text search across patientName and phoneNumber

Returns:

```
json
{
  "data": [
    {
      "_id": "string",
      "patientName": "string",
      "phoneNumber": "string",
      "preferredDate": "string",
      "preferredTime": "string",
      "reason": "string",
    }
  ]
}
```

```

    "preferredContactMethod": "call",
    "source": "whatsapp",
    "isCompleted": false,
    "createdAt": "date",
    "updatedAt": "date",
    "completedAt": "date"
  }
],
"pagination": {
  "page": 1,
  "limit": 20,
  "total": 1,
  "totalPages": 1
}
}

```

Errors:

- 400 for invalid pagination.
- 400 for invalid appointment status.
- 500 for unexpected database errors.

PATCH /api/appointments/:id

Files:

- src/routes/appointments.ts
- src/services/appointments.ts

Purpose:

- Updates only the appointment completion status.

Path parameters:

- id: MongoDB ObjectId string.

Body:

```

json
{
  "isCompleted": true
}

```

Returns:

```

json
{
  "success": true,
  "data": {
    "_id": "string",
    "patientName": "string",
    "phoneNumber": "string",
    "preferredDate": "string",
    "preferredTime": "string",
    "reason": "string",
  }
}

```

```
"preferredContactMethod": "call",
"source": "whatsapp",
"isCompleted": true,
"createdAt": "date",
"updatedAt": "date",
"completedAt": "date"
}
}
```

Rules:

- When isCompleted is true, completedAt is set.
- When isCompleted is false, completedAt is removed.
- No appointment details can be edited by this endpoint.

Errors:

- 400 for invalid Objectid.
- 400 for invalid body or extra fields.
- 404 when appointment is not found.
- 500 for unexpected database errors.

GET /api/users

Files:

- src/routes/users.ts
- src/services/users.ts

Purpose:

- Returns paginated WhatsApp users.

Query parameters:

- page: optional positive integer, default 1
- limit: optional positive integer, default 20, max 100
- search: optional phone number search

Returns:

json

```
{
  "data": [
    {
      "phoneNumber": "string",
      "firstSeenAt": "date",
      "lastActiveAt": "date",
      "totalMessages": 1,
      "createdAt": "date",
      "updatedAt": "date"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 1,
  }
}
```

```
"totalPages": 1
}
}
```

Errors:

- 400 for invalid pagination.
- 500 for unexpected database errors.

GET /api/users/:phoneNumber/messages

Files:

- src/routes/users.ts
- src/services/messages.ts

Purpose:

- Returns paginated conversation history for one WhatsApp phone number.

Path parameters:

- phoneNumber: WhatsApp phone number.

Query parameters:

- page: optional positive integer, default 1
- limit: optional positive integer, default 20, max 100

Returns:

```
json
{
  "data": [
    {
      "phoneNumber": "string",
      "role": "user",
      "content": "string",
      "createdAt": "date"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 1,
    "totalPages": 1
  }
}
```

Errors:

- 400 for invalid pagination.
- 400 for invalid phone number.
- 500 for unexpected database errors.

GET /api/knowledge

Files:

- src/routes/knowledge.ts
- src/services/knowledge.ts

Purpose:

- Reads the current file-based clinic knowledge.

Parameters:

- None.

Returns:

```
json
{
  "content": "markdown content"
}
```

Errors:

- 500 for unexpected filesystem errors.

PUT /api/knowledge

Files:

- src/routes/knowledge.ts
- src/services/knowledge.ts

Purpose:

- Replaces knowledge.md using a temporary file followed by rename.

Body:

```
json
{
  "content": "updated markdown content"
}
```

Validation:

- content must be a string.
- content must not be empty after trimming.
- content must be at most 100000 characters.

Returns:

```
json
{
  "success": true
}
```

Errors:

- 400 for invalid knowledge content.
- 500 for unexpected filesystem errors.

Important Files By Responsibility

- Route registration: `src/routes/index.ts`
- API path constants: `src/constants/api.ts`
- Database collection and index constants: `src/constants/database.ts`
- Pagination constants: `src/constants/pagination.ts`
- Appointment constants: `src/constants/appointments.ts`
- Knowledge constants: `src/constants/knowledge.ts`
- Pagination helper: `src/lib/pagination.ts`
- ObjectId helper: `src/lib/object-id.ts`
- Request body validators: `src/lib/request-body.ts`
- Regex escaping helper: `src/lib/regex.ts`
- WhatsApp parser: `src/parsers/whatsapp.ts`
- WhatsApp sender: `src/services/whatsapp.ts`
- OpenAI service: `src/services/openai.ts`
- Conversation orchestrator: `src/services/conversation.ts`
- Appointment persistence and dashboard operations: `src/services/appointments.ts`
- User persistence and listing: `src/services/users.ts`
- Message persistence and history: `src/services/messages.ts`
- Dashboard stats: `src/services/dashboard.ts`
- Knowledge file service: `src/services/knowledge.ts`